

Enhancement of nondegenerate four-wave mixing based on electromagnetically induced transparency in rubidium atoms

Yong-qing Li* and Min Xiao

Department of Physics, University of Arkansas, Fayetteville, Arkansas 72701

Received February 1, 1996

We report an experimental observation of the enhancement of nondegenerate four-wave mixing (NDFWM) based on electromagnetically induced transparency (EIT) in a lambda-type three-level system of rubidium atoms. We measured both the linear susceptibility $\text{Im } \chi_D^{(1)}$ (absorption) and the third-order nonlinear coefficient $\chi_D^{(3)}$ separately for the NDFWM process at a low atomic density. We found that, owing to the EIT effect, the linear absorption term $\text{Im } \chi_D^{(1)}$ is greatly reduced, while the nonlinear generation term $\chi_D^{(3)}$ is resonantly enhanced, permitting us to observe a significant enhancement of the NDFWM signal in an optically dense medium.
© 1996 Optical Society of America

Recently there has been much interest in electromagnetically induced transparency¹⁻⁴ because of its potential applications in nonlinear-optical processes,^{2,5} lasing without inversion,⁶ pulse matching,⁷ and quantum noise reduction.⁸ Continuous-wave EIT effects have been observed both in a ladder-type three-level system³ and a lambda-type⁴ three-level system in atomic Rb vapor. In particular, quantum interference between the lifetime-broadened dressed states has been observed directly, and the dispersion properties of EIT have been measured.^{3,4} Enhanced nonlinear-optical processes using EIT have also been demonstrated in optically dense media with pulsed laser fields.^{2,5} A high phase-conjugate gain owing to coherent population trapping was reported recently in a four-level double- Λ system of sodium vapor by Hemmer *et al.*⁹

In this Letter we report a clear experimental observation of an enhanced nondegenerate four-wave mixing (NDFWM) process, using the EIT effect in a three-level Λ -type system of Rb atoms with cw diode lasers. The key idea of enhancing nonlinear-optical processes by use of EIT (Ref. 5) is that the absorption of the generated field [related to the linear susceptibility $\text{Im } \chi_D^{(1)}$] can be substantially reduced as a result of the EIT effect while, at the same time, the nonlinear coefficient $\chi_D^{(3)}$ associated with the nonlinear-optical generation process is resonantly enhanced. This leads to a significant enhancement in nonlinear-optical generation in an optically dense medium, in which there would be a large absorption for the generated field without the EIT effect. In our experiment we first measured both the linear susceptibility $\text{Im } \chi_D^{(1)}$ (absorption) and the third-order nonlinear coefficient $\chi_D^{(3)}$ for the NDFWM process under a low-atomic-density (optically thin-medium) condition. The results confirmed the predictions that $\text{Im } \chi_D^{(1)}$ is reduced while $\chi_D^{(3)}$ is resonantly enhanced as a result of the EIT effect.⁵ Then we observed the enhancement in the generated NDFWM signal under high-atomic-density (optically thick-medium) conditions.

Our Λ -type three-level system consists of Rb⁸⁷ atoms in a Doppler-broadened vapor cell. As shown in Fig. 1,

the two hyperfine levels $F_g = 1$ and $F_g = 2$, spaced by 6.8 GHz, of the ground-state $5S_{1/2}$ serve as the two lower states of the Λ system. The excited state $5P_{1/2}$, $F_e = 1$ serves as the common upper state.⁴ The other hyperfine level of the excited state ($5P_{1/2}$, $F_e = 2$) is 814 MHz away (outside the Doppler-broadening linewidth), and its effect can be neglected. The NDFWM is in a forward configuration, i.e., one pumping wave (with frequency ω_2) is on resonance with the $F_g = 2$ to $F_e = 1$ transition, and the other pumping wave ω_1 is far off resonance (with 450-MHz detuning) with the $F_g = 1$ to $F_e = 1$ transition. As the weak probe wave ω_p is tuned to be $\omega_2 - 450$ MHz, a phase-conjugate wave ω_c will be generated at a resonant frequency with the $F_g = 1$ to $F_e = 1$ transition, satisfying the phase-matching condition of $\omega_1 + \omega_2 = \omega_p + \omega_c$. In steady state most atoms will be optically pumped to the ground state $F_g = 1$. If there were no EIT effect, the generated conjugate wave at frequency ω_c would be greatly absorbed by the

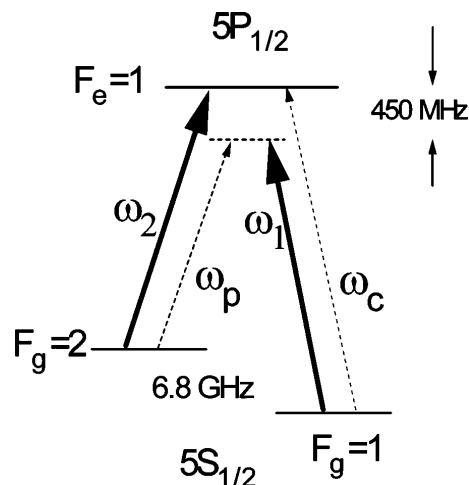


Fig. 1. NDFWM based on EIT in a three-level Λ -system of Rb⁸⁷ atoms. Pumping waves are at frequencies ω_1 and ω_2 , and the probe wave is at frequency ω_p . The phase-conjugate wave is generated at the frequency $\omega_c = \omega_1 + \omega_2 - \omega_p$ and experiences transparency produced by the pumping wave at frequency ω_2 .

dense medium as a result of the resonance absorption. However, the EIT effect in this Λ -type three-level system⁴ causes the atomic medium to be transparent for the generated conjugate wave of frequency ω_c . This permits us to increase the atomic density of the medium and thus increase the total efficiency of the nonlinear-optical generation process, as demonstrated by Hakuta *et al.*² and Harris *et al.*⁵ The propagation equation of the amplitude of a generated conjugate wave in the atomic medium can be expressed by⁵

$$\frac{\partial E_c}{\partial z} + \frac{1}{c} \frac{\partial E_c}{\partial t} = j \frac{\omega N}{2c} \left[\chi_D^{(1)} E_c + \chi_D^{(3)} E_1 E_2 E_p^* \right], \quad (1)$$

where $E_i (i = 1, 2, p, c)$ is the amplitude of the light-wave with frequency ω_i , N is the atomic density of the medium, and $\chi_D^{(1)}$ and $\chi_D^{(3)}$ are the linear and third-order susceptibilities for the $F_g = 1$ to $F_e = 1$ transition owing to a single atom. The first term [proportional to $\chi_D^{(1)}$] on the right-hand side of Eq. (1) describes the linear absorption and dispersion of the medium for the generated conjugate wave, and the second term [proportional to $\chi_D^{(3)}$] describes the nonlinear four-wave mixing process.

The linear susceptibility $\chi_D^{(1)}$ in the steady state can be measured by the standard method^{3,4} in the absence of the pumping wave ω_1 ($E_1 = 0$). The steady-state solution of Eq. (1) in an optically thin medium [$\chi_D^{(1)} NL \ll 1$] is given by $E_c = (j\omega/2c)NL \chi_D^{(3)} E_1 E_2 E_p^*$. Therefore the third-order susceptibility $\chi_D^{(3)}$ can be measured directly from the generated conjugate wave E_c in an atomic system with a low atomic density (optically thin medium).

The experimental setup is shown in Fig. 2. The diode laser, DL1, serves as pumping wave ω_1 with an optical power of 0.3 mW, corresponding to an intensity of 17.9 mW/cm² at the medium. The diode laser, DL2, serves as the pumping wave ω_2 with a typical optical power of 10 mW (corresponding to an intensity of 600 mW/cm²) at the medium. The probe wave ω_p is produced from the sideband modulation of the pumping wave ω_2 by application of a rf signal ($\sim \pm 10$ mV) from a function generator to diode laser DL2.¹⁰ The rf modulation frequency ω_m is fixed at 450 MHz. The sideband intensity is approximately 3×10^{-4} that of the pumping wave carrier (ω_2). Pumping wave ω_2 and probe wave ω_p have cross polarization with respect to that of the pumping wave ω_1 . These three waves pass collinearly through a temperature-controlled Rb vapor cell that is 45 mm long. The generated conjugate wave $\omega_c (= \omega_1 + \omega_m)$, which has the same polarization as that of the pumping wave ω_1 , is detected by the beat signal at the frequency ω_m between the conjugate wave ω_c and the pumping wave ω_1 in a silicon avalanche photodiode (Hamamatsu S2383), which is connected to a spectrum analyzer.

To measure $\text{Im} \chi_D^{(1)}$ and $\chi_D^{(3)}$ in an optically thin medium, we set the Rb cell at room temperature such that the atomic density is approximately $N = 2.6 \times 10^{10} \text{ cm}^{-3}$, at which the maximum absorption coefficient of the $F_g = 1$ to $F_e = 1$ transition is $\sim 8\%$, as shown in Fig. 3(a). When measuring $\text{Im} \chi_D^{(1)}$ we set the amplitude of the rf modulation signal to zero ($E_p = 0$), decrease the intensity of the pumping wave

ω_1 to $\sim 50 \mu\text{W}/\text{cm}^2$ with a neutral-density filter, and scan ω_1 across the atomic resonance. As expected, a reduction in absorption at the resonant frequency is found as a result of the EIT effect. The NDFWM signal $E_c E_1^* + E_c^* E_1 \propto \text{Re} \chi_D^{(3)}$ is recorded by the beat signal between the generated conjugate wave ω_c and the pumping wave ω_1 when ω_1 is tuned across its center frequency of $\omega_{21} - 450 \text{ MHz}$. This yields the third-order susceptibility $\text{Re} \chi_D^{(3)}$, as shown in Fig. 3(b). The main result shown in Fig. 3 is that, owing to the EIT effect, the measured third-order susceptibility $\text{Re} \chi_D^{(3)}$ (the real part) is resonantly

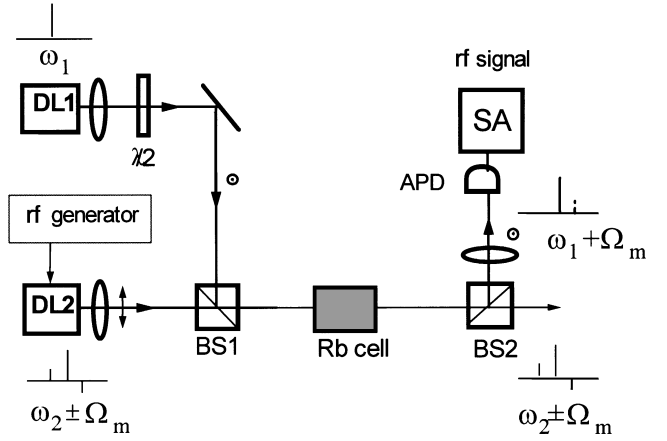


Fig. 2. Experimental setup. BS1, BS2, polarization cube beam splitters; SA, spectrum analyzer; APD, avalanche photodiode; $\lambda/2$, half-wave plate. DL2 is modulated by a rf signal at frequency ω_m in the driven current.

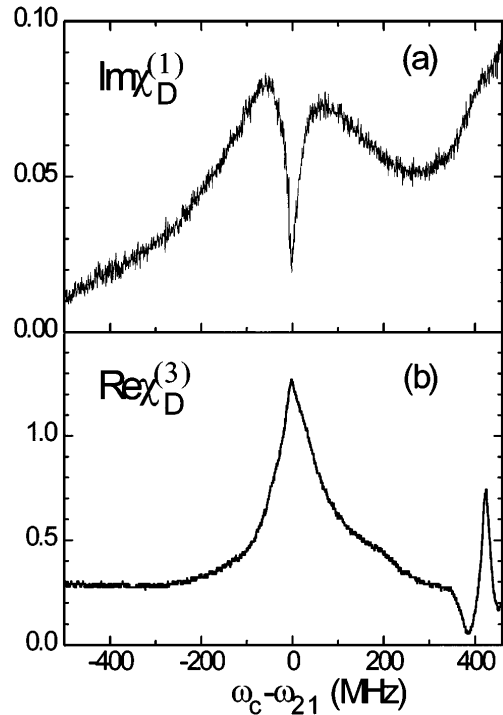


Fig. 3. (a) Measured linear susceptibility (absorption) $\text{Im} \chi_D^{(1)}$ for the phase-conjugate wave versus frequency detuning. (b) Third-order susceptibility $\text{Re} \chi_D^{(3)}$ for the NDFWM in an optically thin medium with an atomic density of $2.6 \times 10^{10} \text{ cm}^{-3}$. Both $\text{Im} \chi_D^{(1)}$ and $\text{Re} \chi_D^{(3)}$ are in arbitrary units.

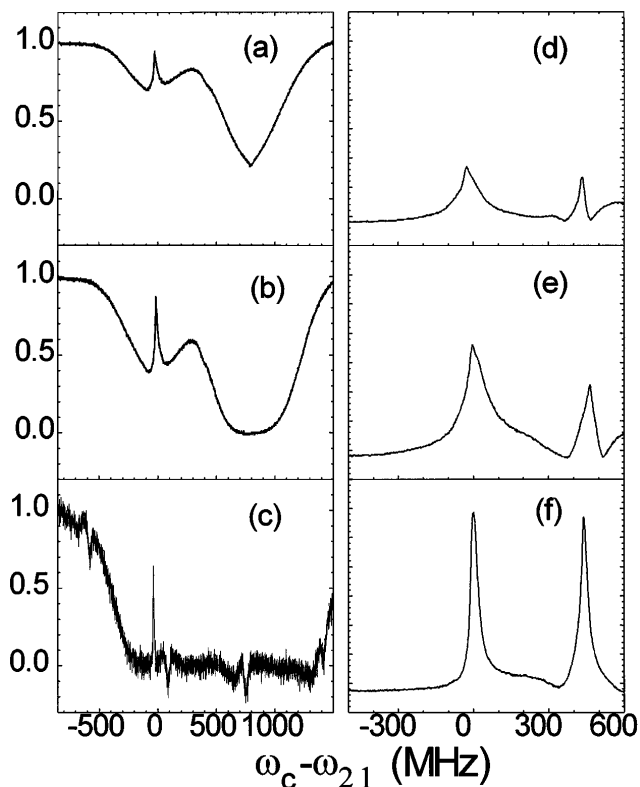


Fig. 4. Normalized transmission of the phase-conjugate wave (left column) and the NDFWM beat signal (right column) at an atomic density of (a) and (d) $9.8 \times 10^{10} \text{ cm}^{-3}$, (b) and (e) $2.0 \times 10^{11} \text{ cm}^{-3}$, (c) and (f) $1.5 \times 10^{12} \text{ cm}^{-3}$.

increased while the absorption $\text{Im } \chi_D^{(1)}$ is reduced at the resonant frequency. The other peak in the NDFWM signal [Fig. 3(b)] is due to the population trapping effect when the pumping frequency ω_1 is tuned to be on resonance with the $F_g = 1$ to $F_e = 1$ transition.

Figure 4 shows the results as we increase the atomic density of the Rb vapor. The left column is the normalized transmission of the conjugate wave, and the right column is the NDFWM beat signal at different atomic densities. The remarkable feature in Fig. 4 is that, as the atomic density is increased, the absorption of the atomic medium for the generated conjugate wave is increased from 25% in Fig. 4(a) to 99.9% in Fig. 4(c). Thus the generated NDFWM signal is decreased at off-resonance frequencies. However, because of the EIT effect, only at the resonant frequency (or within the EIT frequency window) can the generated conjugate wave pass through the dense medium with reduced absorption. Therefore the NDFWM beat signal in Fig. 4(f) is increased in the optically dense medium but has a much narrower linewidth compared with that in the low-density medium.

In conclusion, we have demonstrated a significant enhancement of the nondegenerate four-wave mixing process by using electromagnetically

induced transparency in a Λ -type three-level system of Rb atomic vapor. Both the linear absorption coefficient $\text{Im } \chi_D^{(1)}$ and the third-order susceptibility coefficient $\text{Re } \chi_D^{(3)}$ have been measured in a low-atomic-density medium. In the medium with a high atomic density, the large reduction in absorption owing to the EIT effect leads to a significant enhancement in the total efficiency of the NDFWM process. As a result of the large reduction in absorption (90%) of the generated conjugate wave in our experiment, this enhancement will be useful in generating squeezed light by means of NDFWM (Ref. 11), in which any optical losses (absorption) for the signal light will destroy the squeezing.

We acknowledge funding support from the U.S. Office of Naval Research and the National Science Foundation through grants PHY9221718 and PHY9457905.

*Present address, University of Canberra, Faculty of Information Science & Engineering, P.O. Box 1, Belconnen, ACT 2616, Australia.

References

1. K. J. Boller, A. Imamoglu, and S. E. Harris, Phys. Rev. Lett. **66**, 2593 (1991); J. E. Field, K. H. Hahn, and S. E. Harris, Phys. Rev. Lett. **67**, 3062 (1991).
2. K. Hakuta, L. Marmet, and B. P. Stoicheff, Phys. Rev. Lett. **66**, 596 (1991); G. Z. Zhang, K. Hakuta, and B. P. Stoicheff, Phys. Rev. Lett. **71**, 3099 (1993).
3. Y. Q. Li, S. Z. Jin, M. Xiao, Phys. Rev. A **51**, R1754 (1995); M. Xiao, Y. Q. Li, S. Z. Jin, and J. Gea-Banacloche, Phys. Rev. Lett. **74**, 666 (1995); J. Gea-Banacloche, Y. Q. Li, S. Z. Jin, and M. Xiao, Phys. Rev. A **51**, 576 (1995).
4. Y. Q. Li and M. Xiao, Phys. Rev. A **51**, R2703 (1995); Y. Q. Li and M. Xiao, Phys. Rev. A **51**, 4959 (1995).
5. S. E. Harris, J. E. Field, and A. Imamoglu, Phys. Rev. Lett. **64**, 1107 (1990); M. Jean, G. Y. Yin, J. E. Field, and S. E. Harris, Opt. Lett. **18**, 998 (1993).
6. A. S. Zibrov, M. D. Lukin, D. E. Nikonov, L. Hollberg, M. O. Scully, V. L. Velichangly, and H. G. Robinson, Phys. Rev. Lett. **75**, 1499 (1995).
7. S. E. Harris, Phys. Rev. Lett. **70**, 552 (1993); J. H. Eberly, M. L. Pons, and H. R. Haq, Phys. Rev. Lett. **72**, 56 (1994).
8. K. M. Gheri and D. F. Walls, Phys. Rev. A **49**, 4134 (1994); M. Fleischhauer, Phys. Rev. Lett. **72**, 989 (1994); G. S. Agarwal, Phys. Rev. Lett. **71**, 1351 (1993).
9. P. R. Hemmer, D. P. Katz, J. Donoghue, M. Cronin-Golomb, M. S. Shahriar, and P. Kumar, Opt. Lett. **20**, 982 (1995); J. Donoghue, M. Cronin-Golomb, J. S. Kane, and P. R. Hemmer, Opt. Lett. **16**, 1313 (1991).
10. S. Kobayashi, Y. Yamamoto, M. Ito, and T. Kimura, IEEE J. Quantum Electron. **QE-18**, 582 (1982); S. Saito, O. Nilsson, and Y. Yamamoto, IEEE J. Quantum Electron. **QE-18**, 961 (1982).
11. R. E. Slusher, B. Yurke, P. Grangier, A. LaPorta, D. F. Walls, and M. D. Reid, J. Opt. Soc. Am. B **4**, 1453 (1987).